Can you Erase the Mark of a Criminal Record? Labor Market Impacts of Criminal Record Remediation

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We investigate whether removing a previously-obtained criminal record improves employment outcomes. We estimate the causal impact of criminal record remediation laws that have been widely enacted with the goal of improving employment opportunities for millions of individuals with records. We find consistent evidence that removing an existing record does not improve labor market outcomes, on average. A notable exception is participation in gig work through online platforms, which often screen workers based on their records but not their employment histories. The evidence is consistent with records initially scarring labor market trajectories in a way that is difficult to undo later.
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Millions of people across the United States are charged with crimes each year, with each charge adding additional "marks" to an individual's criminal record.[1] The employment consequences of these records could be substantial. In several surveys, 70-95% of employers state that they perform background checks for most positions (Society for Human Resource Management 2012, HireRight 2015, hr.com 2017, Agan et al. 2024) and many job applications ask about criminal history when permitted (Denver, Pickett and Bushway 2018; Agan and Starr 2018). Consistent with these practices, past research has shown that employment rates are lower for individuals with a felony conviction than the general population (Garin et al. 2022; Mueller-Smith 2015; Looney and Turner 2018) and field experiments have found lower callback rates for applicants with criminal records (Pager 2003; Pager, Bonikowski and Western 2009; Uggen et al. 2014; Agan and Starr 2018). Our own survey also finds that employers state they are less likely to hire individuals with records (Table 1). The magnitudes of the number of people with records and potential impacts on employment rates are so large that they could have potential aggregate implications for the employment rate in the US, particularly for men and communities of color (Looney and Turner 2018; Brame, Bushway and Paternoster 2014; Shannon et al. 2017).[2]

To increase opportunities for individuals with criminal records, jurisdictions across the country have expanded initiatives to expunge or seal criminal records. We refer to these efforts collectively as "record remediation policies." These policies offer a great promise: if records alone are preventing people from accessing employment opportunities, clearing a record can be done with relatively little cost, and in some jurisdictions can be done wholesale.

In this article, we study the labor market impacts of three major criminal record remediation policies in the United States that exogenously change or limit the availability of criminal history information reported in employment background checks. Two of these policies focus on remediating non-conviction records, and one remediates conviction records. All three remediate both felony and misdemeanor level charges. While much of the past research focuses on felony convictions, most charges filed are misdemeanors (Mayson and Stevenson, 2020) and conditional on charges being filed by the prosecutor, many cases will end up in a non-conviction, usually via dismissal.[3] Even these lower-level charges and

[1] Nearly 97 million individuals have a criminal history record stored in the Federal Bureau of Investigation's (FBI) Interstate Identification Index (III) based on records maintained by the FBI and state criminal record repositories, covering over one-third of the adult working-age population (Goggins and DeBacco 2020).

[2] Having a criminal record can also affect access to housing and public benefits (Evans and Porter 2015; Leasure and Martin 2017; Yang 2017; Tuttle 2019), which can further affect employment prospects.

[3] According to the Court Statistics Project, 77% of cases in state courts in 2022 were misdemeanors (https://www.courtstatistics.org/court-statistics/interactive-caseload-data-displays/csp-stat). In 2009 in a random sample of cases from the 75% largest counties in the United States, 25% of felony charges were dismissed, another 10% were acquitted or given diversion (Reaves 2013). Among misdemeanor charges in a sample of 8 jurisdictions that spanned from rural to urban, between 28 - 73% of charges ended in a dismissal or diversion depending on the jurisdiction (Mayson and Stevenson 2020). Among misdemeanor charges in Suffolk County, MA, 74% ended in a non-conviction (Agan, Doleac and

non-conviction outcomes can show up on criminal background checks, potentially reducing employment opportunities for people with records.

To examine employment trajectories before and after criminal legal contact and estimate the labor market impacts of the criminal record remediation policies we study, we link publicly-available individual-level criminal records data from four jurisdictions—Maryland, New Jersey, Pennsylvania, and Bexar County, Texas—to administrative tax records at the Internal Revenue Service (IRS).

We begin by providing evidence of large and persistent drops in formal employment at the time of both misdemeanor and felony charges. These patterns are observed for convictions and also, perhaps more surprisingly, for non-convictions. Though these drops may coincide with other shocks happening at the time of the criminal events, these findings are consistent with previous data from audit studies that has shown that employers react negatively to felony convictions and also to misdemeanor arrests (Pager 2003; Uggen et al. 2014; Agan and Starr 2018). Further, a survey of hiring decision-makers at firms that we conducted supports the interpretation that these patterns at least partially reflect the negative impact of having a record. Hiring professionals report markedly reduced likelihood of hiring someone with a drug or theft charge, even if it resulted in a non-conviction.

We then study whether criminal record remediation can increase employment. The first policy we study is the federal Fair Credit Reporting Act (FCRA) which prohibits reporting of criminal charges that did not lead to a conviction (mainly dismissals) after seven years for jobs that pay less than $75,000 a year. In addition, several states have a stronger version of the law that applies to convictions, in particular Maryland, which is one jurisdiction in our study.[4] These laws apply to all consumer reporting agencies (CRAs), which include the major credit bureaus and employment background check companies. We examine the evolution of labor market outcomes after the last criminal charge or conviction with a focus on the sharp seven-year change in reporting, comparing the outcomes for individuals before and after the record is removed. In our first analysis, we focus on individuals whose last criminal charge is a non-conviction and who also have no other convictions in that jurisdiction, such that FCRA gives these individuals a clear criminal record on CRA-run background checks seven years after the last event. Using the same design, we examine reporting prohibitions for convictions, contrasting the experience of Maryland—a state that prohibits the reporting of convictions after seven years for lower earning workers— with other states with no such restrictions.

We also study Pennsylvania's Clean Slate Law of 2018, which legislated automated sealing of *all* non-convictions. Our data allow us to analyze the impact of these sealings on charges that were at least 18 months old using a difference-in-differences research design, comparing outcomes of individuals who only had

Harvey 2023).

[4]For a full list of different FCRA rules by state, see Appendix D.

non-convictions on their record and had their non-conviction sealed to a comparison group of individuals with non-convictions who did not have their record sealed because they had unpaid fines and fees and thus were not eligible for the first round of automated sealing.

Our findings are consistent across the different criminal record remediation policies we study and across jurisdictions. In all but a few notable cases, we find little evidence that clearing a record improves labor market outcomes for the average affected individual. We find no evidence that removal of non-conviction records from criminal background checks at seven years under FCRA increases the likelihood of having any formal employment income for those whose records are completely clear seven years after their last non-conviction event. This finding holds across almost every employment-related outcome. Similarly, using variation in Maryland, we fail to find a detectable improvement in average employment rates seven years after an individual's last conviction event, when their record becomes completely clear. These results are further supported by the analysis of the Pennsylvania Clean Slate Law, where we find no effect of having non-conviction records 18 months or older automatically sealed on employment outcomes amongst those with only non-convictions on their record.

Ultimately, our reading of the totality of the evidence is that existing record remediation policies have limited effects on employment, at least on average. Why, then, are there such persistent impacts of having a criminal record, even for a misdemeanor or non-conviction? Our hypothesis is a scarring mechanism: records initially create gaps in employment or limit access to opportunities with job ladders or long-run employment potential which, in turn, lead to subsequent adverse outcomes separate of the record itself.[5] A notable exception to our general null finding is evidence that record remediation policies increase the rate of electronically-mediated gig platform work, albeit from a very low base.[6] In the case of platform-based gig work, such as ride-sharing, all applicants go through a background check but face few other barriers to participating as a worker. While jobless spells and intermittent employment caused by having a record may impact consideration for many jobs, this type of labor market scarring has little scope to impact an applicant's ability to find work on a gig platform, so long as they pass the background check. And thus the finding of a positive effect there is perhaps indicative that labor market scarring is partially driving our null results.

Our findings indicate that scarring from records can occur quickly, consistent with recent work on the downstream effects of initial misdemeanor charges (Agan, Doleac and Harvey, 2023). For example, we find that even record remediation occurring as early as 18-24 months following a non-conviction event, as in our PA Clean Slate analysis, appears to be too late to undo scarring.

[5]The labor literature shows that displaced workers tend to experience persistent earnings losses (Jacobson, LaLonde and Sullivan 1993; Lachowska, Mas and Woodbury 2020). Kroft, Lange and Notowidigdo (2013) shows that employers are less likely to callback applicants with longer unemployment spells.

[6]We do not find evidence of effects on other types of self-employment work.

These results mirror results from work studying the impact of sealing bankruptcy flags and clearing medical debt. Using a similar methodological approach, Goldsmith-Pinkham et al. (2017) examine the effects of removing bankruptcy flags on individual outcomes and find limited effects on earnings and employment. Other work using RCTs, such as Kluender et al. (2024), find that medical debt relief, when relieved more than a year after the initial debt was acquired, has no impact on credit outcomes, financial distress, or mental health. In a different experiment when medical debt was relieved immediately before patients left the hospital, there were increases in healthcare use after relief including increases in detection of diabetes and depression (Adams et al. 2022).

Of course, criminal record remediation policies may have beneficial impacts on individuals that are not reflected in the formal labor market earnings we observe in IRS tax records. Remediation of records could directly impact access to housing, civic engagement, quality of life, and other policy-relevant outcomes. At the same time, our analysis suggests that if reintegration and labor market participation are primary objectives, existing policies are not achieving these goals. Our findings may help explain why recent evaluations of Ban-the-Box policies failed to find improvements in labor market outcomes for individuals with records (Rose 2021).

Our work is closely related to several literatures. One literature studies the impact of criminal record remediation laws by studying cases where individuals file their own petitions for record remediation where available (e.g. via expungement or sealing of older or low-level records). This work documents a substantial "uptake gap": a very small proportion of individuals that are eligible for this type of remediation take it up (Chien 2020). Previous work that has studied the labor market impacts of expungement amongst those who choose to petition has found a positive trend in employment after the petition-based expungement (Selbin, McCrary and Epstein 2018; Prescott and Starr 2019). There may be significant selection into who decides to take up this treatment; our study builds on this prior work by studying settings where individuals do not select into treatment.

Our study is also closely related to the literature studying the interplay between crime, criminal records, and employment more generally. For example, previous work using tax data has found that formal employment prospects for individuals who have been released from prison are poor both before and after the period of incarceration (Looney and Turner 2018) and conviction (Rose 2021). Quasi-experimental evidence on the employment impacts of incarceration yields mixed results (Kling 2006; Mueller-Smith 2015; Bhuller et al. 2020; Garin et al. 2022). This work tends to focus on individuals who have been convicted.

There has been more limited work on labor market impacts of criminal justice events before the conviction stage, with Freeman (1991), Grogger (1995), Dobbie, Goldin and Yang (2018), and Mueller-Smith and Schnepel (2021) being notable examples. Very few, if any papers, have studied employment trajectories around non-convictions. Grogger (1995) finds that the impact of an arrest on employment is small and not persistent in the 1980s in California. Not all arrests will lead to

charges or convictions, and interestingly, since 1975, California is one of the few states that prohibits the reporting of non-convictions which could dampen labor market impacts of an arrest. Dobbie, Goldin and Yang (2018) finds that pre-trial detention versus release is associated with decreased employment, but does not focus on non-convictions or convicitons per se. Agan, Doleac and Harvey (2023) studies the impact of a misdemeanor charge versus an arrest that did not lead to a charge on recidivism; they find that being charged with a misdemeanor (of which 75% end up dismissed) significantly increases the probability of recidivism over being arrested with no charge for those on the margin of being charged by Suffolk County, MA prosecutors. They posit a channel that runs through the "mark of a criminal record" but do not have employment outcomes they could leverage to directly assess this hypothesis. Rose (2021) presents similar event studies to ours, showing the trajectory of employment around a conviction (for those that led to incarceration and those that did not), but does not consider the trajectory of employment around the time of a non-conviction. In a different policy context, Mueller-Smith and Schnepel (2021) compares outcomes for defendants all of whom are charged with felonies, but some of whom get deferred adjudication versus a conviction, and find that access to deferred adjudication increases employment. While our study cannot uncover the causal impact of a non-conviction, it does illustrate the trajectory of employment around the time of a non-conviction event, an important criminal justice event, and is an important first step in understanding the potential causal impact of non-convictions.

In a different setting, Agan et al. (2024) use experimental and quasi-experimental variation to study employment outcomes for defendants who get their existing felony convictions retroactively reduced to misdemeanor convictions. That paper similarly finds null employment effects for individuals who did not select into the treatment (and positive effects for people who did, consistent with the previous petition-based literature) in one county in California. In this paper, we study remediation amongst a subset of the population for which remediation policies result in a *completely clear* background check and in a broader set of jurisdictions. If employers react negatively to the existence of anything on a background check, then clearing it completely could have different impacts than simply reducing its severity.

Researchers have previously studied two other policies that limit criminal record access in different jurisdictions and policy contexts. Dasgupta, Ghimire and Plum (2025) study the enactment of New Zealand's clean slate regulation, which allowed automatic sealing of criminal convictions for individuals with no further convictions within seven years. In that sense, the New Zealand law most closely mimics Maryland's version of FCRA and our analysis of that policy. Consistent with our findings in the U.S. context, Dasgupta, Ghimire and Plum (2025) find null effects on employment outcomes at the seven-year mark. We build on this work by studying non-convictions, which can get reported on background checks in the US, in a different setting and country where criminal background checks are even

more prevalent. Jackson, Sullivan and Zhao (2017) study the introduction of several criminal records policies in Massachusetts, including a ban the box policy and the introduction of a new background check database that provided an alternative to CRA background checks and also limited access to older records. They similarly find no evidence of employment benefits. However, the database also only limited how convictions were reported, and did not report non-convictions. In addition, employers are not required to use the new background check database, which only offers access to MA criminal histories. Our work is able to consider non-convictions and study a context where the policy universally applies.

Finally, our work relates to an experimental literature that has consistently found lower callback rates for those with records, including misdemeanor arrests (Pager 2003; Uggen et al. 2014; Agan and Starr 2018). And beyond standard employment outcomes, we also build on earlier work of Collins et al. (2018) to examine whether criminal records may push individuals out of the traditional workforce and into alternative work arrangements such as platform-based gig work and other independent contracting or self-employment.

The article is organized as follows. Section I discusses the policies we study, Section II discusses the data, and Section III describes survey and observational results that document the penalty associated with criminal records. Sections IV and V describe our empirical specifications and results. Section VI offers some concluding remarks.

I. Criminal Record Disclosure Under Reporting Laws

In this section, we discuss the three policies that are central to our analysis on the impacts of the "extensive margin" of record reporting: the federal Fair Credit Reporting Act that sets a seven-year time limit for reporting of non-conviction criminal records by CRAs, the stronger law in Maryland that also applies to convictions at the seven-year mark, and the pioneering clean slate law in Pennsylvania.[7] Appendix Figure A.2 shows a map of which states in the U.S. have clean slate policies or stronger FCRA requirements as of 2024.

A. Background

FCRA: The federal FCRA governs the type of information reported by CRAs to potential employers.[8] FCRA only applies to criminal background checks performed by CRAs (which includes major credit bureaus and also employment background check companies). A firm that performs an in-house criminal background

[7]Appendix Figure A.1 shows an example actual background check from a CRA which includes a dismissed charge.

[8]There are two types of background checks an employer can choose to run: fingerprint checks and name searches. In a fingerprint-based search, arrest records will appear regardless of whether the arrest leads to a formal charge. A name-based search will turn up court charges even if they did not lead to conviction, although is unlikely to uncover arrests that did not lead to a court charge. Name-based searches are more common for most mainstream types of employment.

check is not subject to FCRA requirements. Relevant to our context, under FCRA, non-convictions are only reportable for seven years for jobs with annual expected salary <$75,000; in contrast, convictions are always reportable regardless of the age of conviction. The seven-year clock for non-convictions starts at the date the charge is filed.[9] We use the seven-year rule under FCRA to estimate how employment changes after non-convictions are cleared from individuals' records. In particular, under this rule, an individual's criminal history should be completely clear seven years after the last criminal history event/charge among individuals who have no convictions on record. This feature of FCRA allows for an event-study design where individuals do not select into the event in the relevant time horizon for estimation. Our approach is in the spirit of Goldsmith-Pinkham et al. (2017) who examine the effects of removing bankruptcy flags on individual outcomes.

Our survey of hiring professionals further discussed in Section III—in particular the responses in Appendix Table A.1—confirms that this is a relevant policy for understanding the role of records on employment outcomes. There is considerable evidence that background checks are overwhelmingly done through CRAs. Amongst the respondents whose most recent firm did criminal background checks for entry-level positions (70% of the sample), 85% of criminal background checks for entry-level positions were performed by a CRA that would be obligated to follow the look-back provisions of FCRA for reporting non-convictions.[10] While it is plausible that employers may be doing other types of informal searching on top of CRA checks (Lageson 2020), given the myriad laws surrounding criminal background checks and the large proportion of employers using CRAs, even for entry-level jobs, it seems likely that FCRA's reporting requirements directly affect the information visible to employers.

To more directly verify CRA compliance with FCRA, we obtained data on all background checks performed by a large CRA for 288 firms covering more than 11 million charges. In Figure 1, we plot the distribution of time since criminal event for conviction charges and non-convictions that show up on criminal background checks from this CRA. As seen in the figure, no non-conviction charge is reported after seven years, consistent with the reporting requirements under FCRA. Since FCRA does not regulate the reporting of convictions, most convictions that are older than seven years are reported. However, there is a marked discontinuity in the frequency of convictions reported at seven years because of analogous laws in nine states that further restrict the reportability of criminal convictions at seven years.[11]

[9] In the context of recent litigation, courts have upheld the Federal Trade Commission and Consumer Financial Protection Bureau's interpretation that the FCRA look-back window for non-convictions starts at the date of filing and NOT the date of dismissal.

[10] 78% said this check was performed by an external CRA and they could name the CRA (and we could verify that it was, in fact, a CRA). Only 9% of respondents said they performed the background check "'in-house" by searching court records themselves. The remaining 6% reporting using fingerprint checks or other official state record searching programs such as CORI in Massachusetts.

[11] See https://help.checkr.com/hc/en-us/articles/360001100688-What-are-state-salary-excep

Maryland Credit Reporting Law: We also study a law in Maryland, passed in 1976, which prohibits employers from requesting arrest or conviction records that are more than seven years old (starting from the date of disposition) for any job that pays less than $20,000 per annum.[12] While this income threshold is low, it binds for the majority of individuals with criminal records. In our administrative tax data, in Maryland, the average earnings for individuals matched to criminal records data and who are working is $14,000 per year ($7,000 per year median).[13] These numbers suggest that part-time jobs and likely even full-time jobs will be largely under this threshold in our population of individuals with records. In addition, our conversations with a major CRA indicate that the agency's default policy is to not provide prior convictions more than seven years old from date of disposition to any employer in Maryland, regardless of the position's expected annual salary, and the vast majority of employers do not opt out of this default.

Pennsylvania Clean Slate Law: The last policy we study is Pennsylvania's Clean Slate Law of 2018 (Act 56 of 2018), which legislated automated sealing of *all* non-convictions and some older low-level convictions. The Clean Slate law automated the sealing of non-conviction records with no waiting period, as well as certain low-level conviction records after ten years. Eligible records were sealed between June 2019 and June 2020 and the law has since resulted in nearly 40 million criminal records sealed for over 1.2 million individuals. Under this law, these records are automatically shielded from the vast majority of employers, landlords, schools, and the general public, but are still accessible to law enforcement and judicial officers. To estimate the effects of this policy we employ a differences-in-differences design. When the law was initially passed, individuals who owed fines and fees to the court at the time of the initial set of sealings were not (initially) eligible.[14] We therefore compare individuals with eligible records with and without fines and fees, from before the change to the year after the policy in 2021.

II. Data

To analyze the evolution of employment around criminal charges and the impact of criminal record remediation on labor market outcomes, we merge publicly-available criminal records data from several jurisdictions to administrative data from the IRS. In this section, we describe the administrative criminal records and tax data we use.

Our administrative court records come from Maryland, New Jersey, Pennsylvania, Maryland, and Bexar County, Texas. They include all misdemeanor and felony charges filed with the court within the relevant time period, with the ex-

tions-.

[12]MD. CODE. ANN., COM. LAW §14-1203(b)(3) (2010).

[13]Over our sample period of 1999 to 2018, $20,000 is approximately the median annual labor income for employed men without college education in Maryland according to the March Current Population Survey.

[14]PA subsequently eliminated court fees and fines as a barrier to sealing so long as restitution is paid.

ception of our New Jersey data, which just contain information on Superior Court cases (equivalent to felonies in other jurisdictions). In addition, the data include the date of each charge, the disposition date, and ultimate outcome (dismissed, convicted, etc.), as well as other defendant and case characteristics. Further details on these data, including how we match defendants across cases and how we categorize charges are available in Appendix C.[15]

Bexar County, Texas. — The data from Bexar County include all charges filed between 1975 and 2018.[16]

Maryland. — The Maryland criminal records were drawn from several tables hosted on the Maryland Volunteer Lawyers Service (MVLS) database. The Maryland data cover all charges filed between 1990 and 2018.

New Jersey. — We obtained New Jersey court data from the Superior Court of New Jersey via a Public Access Information Request. We only have data for "indictable offenses" which closely align with what would be felonies in other states, and we use the term felonies to describe them. These files contain records from January 1, 1980, to May 30, 2018.

Pennsylvania. — We obtained Pennsylvania court data from the Administrative Office of Pennsylvania Courts (AOPC) via a Public Access request. The data cover all cases in the Magisterial District Court system (which handles misdemeanors) and Courts of Common Pleas system (which handles felonies) filed between May 2008 and April 2018. Using data from the Philadelphia District Attorney's Office, we are able to identify records that were sealed by PA Clean Slate between June 2019 and June 2020. Our estimates indicate that 57% of the nearly 7.5 million non-convictions charges in our AOPC data were sealed, with the ones that were not sealed due to individuals still owing fines and fees at the time of the initial Clean Slate implementation.[17]

[15] Appendix Figure A.3 shows how these jurisdictions compare to the distribution of other states and the U.S. average on crime rates and labor market conditions. These jurisdictions tend to have lower unemployment rates than the average of the U.S. but are not outliers in the distribution of states. Pennsylvania and New Jersey have lower property crime rates than the US average, Texas has a higher rate than the US average, and Maryland is very similar. On Violent crimes Maryland has more than the US average but the other jurisdictions are similar.

[16] The data are publicly available for download via the court website, which began releasing them as part of an effort to make court records more accessible. Full detailing of this data can be found in Agan, Freedman and Owens (2021) and Freedman, Owens and Bohn (2018).

[17] Our research team along with a team the Philadelphia DA's office looked up a small, random subset of case information online for non-convictions that were not sealed between June 2019 and June 2020 and were able to confirm that for that subset, each one owed fines and fees. In October 2020 a new law was passed eliminating court fines and fees as a barrier to sealing remedy under Clean Slate and these additional cases were sealed in late 2021.

A. IRS Employment Outcomes

We draw our outcomes from the universe of IRS tax filings, which are linked to the individual-level court record data. The IRS data includes anyone who has ever filed an individual tax return or has had income reported to the IRS on an information return.

We construct main employment and tax filing outcomes standard in the literature using administrative tax returns. We draw information on formal sector wage and salary earnings and employment from W-2 returns issued by employers for each employee in each year. Importantly, W-2 returns are sent by employers to the IRS irrespective of whether and how the employee files their own individual tax return. Individuals with no W-2s or self-reported income in any particular year are assumed to have had no earnings in that year. We examine whether individuals have any wage and salary earnings as well as whether earnings exceed specified threshold levels. We supplement these earnings records with gross payments to non-employee independent contractors and online platform "gig" workers reported by firms on 1099-MISC and 1099K forms. We construct these earnings measures following the methodology in Collins et al. (2018).[18] W-2 based outcomes are available from 1999 to 2019 and 1099-based outcomes are available from 2000 to 2019, with the exception of platform-mediated gig work which only became widespread enough to measure separately beginning in 2012.

We also examine outcomes directly reported to the IRS by individual taxpayers themselves on Form 1040 tax returns. These include total gross self-employment revenues and net profits after expenses reported on Schedule C, with self-employment net profits further broken out for each individual within the filing unit (with net profits of at least $433) on Schedule SE.[19]

The IRS data are linked to the criminal records data using individual name, date of birth, and geographic location. We were able to match 86%, 73%, 81%, and 91% of the data in Bexar, Maryland, New Jersey, and Pennsylvania, respectively. Further details on our matching algorithm to IRS records can be found in Appendix F. For each jurisdiction, our match rate is comparable or higher than past work matching to IRS or Unemployment Insurance data (e.g. Dobbie, Goldin and Yang 2018; Travis, Western and Redburn 2014). Appendix F also compares characteristics of matched and non-matched individuals and the distribution of last criminal history events in the full versus matched sample for our FCRA and

[18] Unlike payments to independent contractors on 1099-MISC, which are subject to a $600 threshold, online platform economy earnings reported on 1099-K are subject to a much higher $20,000 threshold. Garin et al. (2023) reports that while that most major platforms issued 1099-Ks to all platform participants through 2016 regardless of the earnings level, several large platforms announced that they would adhere to the higher thresholds beginning in 2017. That study estimates that roughly one-third of online platform workers in 2018 were missing in the federal 1099 data because they had gig earnings below the higher 1099-K reporting threshold.

[19] Unlike information returns reported by third-parties, individual tax reporting on 1040 forms may be impacted by taxpayer non-compliance or strategic reporting. Nonetheless, reporting behaviors such as filing any 1040 return are of interest to understand how criminal record remediation impacts tax compliance.

Maryland analysis.

B. *Measuring First and Last Events in the Charge Data*

Within each of the jurisdiction-level datasets, we focus on estimating the trajectory employment around first (known) criminal justice events within that jurisdiction (Section III), or when a policy removes criminal history information after a last event (Section IV). First and last events are measured within the same jurisdiction and within the time frame in the available data. To reduce measurement error associated with measuring past convictions, we limit the sample to individuals who were 18 or older as of the earliest year of data available in the respective jurisdiction to ensure that we can accurately measure their adult criminal record. In theory, cross-jurisdiction charges or convictions could cause measurement error in our estimates. Given that some of our jurisdictions are geographically close, and thus may experience more inter-state crime flows, we checked within our data for matches on first name, last name, and date of birth in a different jurisdiction. We found that only 1.36% of our observations had a match in a different jurisdiction. As a result, we believe this issue is not greatly impacting our estimates and proceed with a within-jurisdiction analysis.

III. Evidence on the Penalty Associated with Criminal Records

Strong quantitative evidence on the informational content of criminal records in employment decisions comes from audit studies (Uggen et al. 2014, Pager 2003, Agan and Starr 2018). These studies have consistently found a causal penalty in callback rates for an applicant reporting a record, both misdemeanor and felony. For example, Uggen et al. (2014) find that employers are four percentage points (10-14% depending on race) less likely to call back individuals reporting a misdemeanor arrest that did not lead to a conviction; this penalty is substantial but smaller than the penalty for reporting felony convictions (Pager 2003; Agan and Starr 2018).[20] Audit studies are limited, however, because they only rely on information entered on applications and the outcome is limited to callbacks.

In this section, we take two additional approaches to documenting whether and how criminal records may harm employment outcomes: (1) a survey of hiring professionals, and (2) documenting changes in employment and earnings around the time of criminal charges.

A. *Survey Evidence*

To learn more about the role of criminal record background checks in hiring decisions, we conducted two surveys of hiring professionals. The first survey

[20] Employers were 50% less likely to callback an applicant with a felony conviction in Pager (2003) and 37% less likely for the felony conviction in Agan and Starr (2018).

(N=808) took place in May 2021 and asked about practices related to criminal background checks at the individual's most recent firm for entry-level positions. The second survey (N=1003) took place between September to October 2021 and presented hiring scenarios to a different group of hiring professionals who reported having recent hiring experience in the United States.[21]

Survey results for the first survey are presented in Appendix Table A.1. 70% of respondents reported that their firm did criminal background checks for entry-level positions (and 7% were unsure), and of these, 59% (N=383) reported being knowledgeable about background check procedures. Importantly for our FCRA analysis, we find that of those who perform background checks, 78-85% report that those checks are performed by CRAs, which are regulated under that law.[22]

In the second survey, we presented a vignette that described a hiring scenario for an entry-level position at the respondent's most recent firm. Specifically, we presented the following scenario:

> *"You intend to hire a candidate for an open entry-level position at the most recent firm at which you had hiring experience. Through the hiring process, you decide that this candidate is well qualified for the position. You are ready to extend an offer to the candidate. However, you learn that the candidate was charged with [felony/misdemeanor] [drug possession/theft] X years ago and was [convicted/not convicted]. How likely are you to recommend that the company hire the candidate?"*

We randomly assigned respondents "felony" or "misdemeanor," "drug possession" or "theft," and "convicted" or "not convicted." The number "X" varies from 1 to 10 for each respondent.

These results from the second survey are reported in Table 1. Applicants with drug possession and theft charges are substantially penalized in hiring, even if they have not been convicted of the charge. Only 60 percent of respondents state that they would hire someone with a misdemeanor drug non-conviction in the previous 3 years. This rate decreases to 49 percent for a recent drug felony non-conviction. For theft offenses, the corresponding rates are 45 percent and 42 percent. Clearly, even a non-conviction is viewed quite negatively by these respondents. Convictions are penalized even further, and moreso for felonies than misdemeanors.

Table 1 also shows that the penalty employers assign to convictions and non-convictions diminishes with the time elapsed since the event. In our survey, depending on the charge, 15 to 25 percent of hiring professionals report that they would pass on qualified applicants with non-convictions for drug and theft charges at year seven, compared to 15 to 35 percent who report they would do

[21] The surveys were conducted using Prolific, which has compared favorably to other platforms for soliciting survey respondents (Peer et al., 2017). See Appendix B for details on the survey sample and its representativeness.

[22] 7% are performed by an external agency, but the respondent could not name the agency so we could not verify it was a CRA.

so for applications with convictions. This finding, which is relevant for all of the policies we consider, provides an additional reason why the expected effect of having a record sealed or reduced may vary with time since charge.[23]

B. *Employment Around a New Charge*

How does employment evolve around the time of criminal legal charge? We examine this in our linked microdata sample focusing first on the (known) first-event for an individual in our sample within each jurisdiction. We then also study how employment evolves around the charge that is the latest observed event for an individual with records to ensure that changes in employment outcomes after the event are not solely due to additional charges or convictions.

We document the change in earnings around these events using an event-study framework.[24] We estimate the model:

$$y_{it} = \sum_k \beta_k \mathbb{1}\{E_i = t + k\} + \alpha_i + \alpha_{a(i,t)} + \alpha_t + \epsilon_{it} \tag{1}$$

We do this separately for convictions and non-convictions and for felonies and misdemeanors. We assign the charge date as the event date for non-convictions and assign the disposition date of the last criminal history event for that defendant in a particular jurisdiction for convictions. α_i is an individual fixed-effect, $\alpha_{a(i,t)}$ are age fixed effects, and α_t is a year fixed effect. We bin endpoints for the period more than 5 years prior and 10 or more years post event. For the non-conviction sample, we limit the latest non-conviction charge to individuals with no earlier conviction on record in the same jurisdiction within our data. The estimates are shown separately for each jurisdiction for which we have complete criminal records data for a sufficiently long panel: Bexar County, TX, Maryland, Pennsylvania, and New Jersey (for felonies only). We present these estimates using data from 2000 to 2020, restricted to "last events" that occur between 2003 to 2011. Importantly, we caution that these event study estimates may not reflect causal estimates of new charges if there are correlated shocks at the time of the new charge.

We present summary statistics for this sample in Appendix Tables A.2-A.3, broken out separately by last events which are non-convictions and convictions,

[23] These survey findings are largely consistent with the past literature. For example, using employer surveys administered in various large metropolitan areas in the 1990s to early 2000s, Holzer, Raphael and Stoll (2003) find that employers are much more averse to hiring ex-offenders than they are towards any other disadvantaged group, and that employer criminal background checks are increasingly used. On an online labor platform that third-party businesses use to connect with workers seeking short-term jobs, Cullen, Dobbie and Hoffman (2023) employ an incentive-compatible survey and similarly find that employers are reluctant to work with people with criminal records (only 39% willing), and this reluctance holds across both misdemeanor and felony convictions, although demand is higher for those with a misdemeanor (e.g. 51% willing to work with a worker with a drug-related misdemeanor, only 27% with a drug-related felony). They did not explicitly ask about non-convictions.

[24] Rose (2021) also uses an event-study approach to estimate the effect of convictions on labor market outcomes.

and by last events that are felony and misdemeanors. We see that two years before the first criminal history event, the employment rates of those with misdemeanor charges, regardless of conviction status, are quite similar at around 77%. Among those whose first event is a felony, pre-period employment rates of those who are convicted are slightly lower (69% compared with 72%). Looking at the last criminal event, those whose last event is a non-conviction with no other convictions had baseline employment rates of 72 or 78% depending on whether the last charge was a felony or misdemeanor, respectively. These rates are quite similar to those around the first event. For those whose latest event was a conviction, baseline rates of employment are substantially lower at 60 or 69% for felony and misdemeanor, respectively.

Figure 2 plots the β_k coefficients for having any wages relative to two years prior to the *first* criminal legal event for an individual within that jurisdiction. For all types of criminal legal contact, employment falls after an initial charge and remains low persistently even 6 years after the event, even when the defendant was not convicted. After an initial criminal legal charge that leads to a felony conviction, employment rates are lower by between 10 and 20 pp depending on the jurisdiction, which represents an 13-26% decline, felony non-convictions are associated with a persistent is 5-8pp decline in employment (8-11%); misdemeanor convictions is 6-9pp (7-11%); by 6 years out after misdemeanor non-conviction is 6pp (a 7-8% decline). These declines in employment are large and importantly remain persistent 6 years after the initial criminal charge. We examine other types of labor earnings reported to the IRS in Appendix Figure A.4 and find no evidence of offsetting increases in independent contract work or other self-employment around an initial charge.

Some of the persistence in the overall decline could come from subsequent criminal legal contact.[25] Figure 3 plots the β_k coefficients for having any wages relative to two years prior to the *last* criminal event. We still see that formal labor sector engagement falls sharply and persistently around the time of a criminal charge, even those that did not lead to conviction. For felony convictions there is some evidence that this is a transitory decline that mimics incapacitation in prison—individuals employment probabilities fall 7-20pp in the year after the conviction and by 6 years have returned closer to pre-conviction baseline. For many people with felony convictions, their last event was preceded by other criminal legal charges—so they are returning to an already depressed baseline from their *first* charge. For felony non-convictions, employment is 3.5 to 5.6 percentage points (4.6-7.9%) lower two years from the last charge. These declines are persistent and only exhibit a limited rebound. For misdemeanor non-convictions, the share of the group with any earnings is 1.3-1.8 percentage points (1.7-2.2%) lower two years following the charge, with no detectable reversion to pre-charge levels.

Figure 4 presents changes in the probability of being employed in a specific

[25] Agan, Doleac and Harvey (2023) shows that initial misdemeanor charges, even when they do not lead to a conviction, cause large increases in recidivism for marginal defendants.

sector around the time of an initial charge (Appendix Figure A.5 does so for last charges). Both non-convictions and convictions appear associated with reallocation in employment across sectors. Around the first nonconviction we see declines in manufacturing, finance, real-estate and health care employment. These sectors with relative declines, particularly finance, real-estate, health care and education tend also to be sectors with occupational licensing requirements. Conversely, following a non-conviction charge we see rises in construction, professional services, temp agencies, and accommodation/food employment. The relative changes in sectors for first convictions largely mirror non-convictions, but there is mostly an absolute decline across all sectors due to lower overall employment probabilities. Nevertheless, there is even an absolute increase in the rate of employment in temporary help agencies. Temporary help agencies appear to serve as a stopgap for people experiencing these adverse events.

When interpreting the overall patterns it is important to note that factors other than criminal records, such as addiction and mental health issues, plausibly contribute to adverse outcomes after criminal charges. In the case of felony convictions, there are clearly incapacitation effects, though these are unlikely for misdemeanors and non-convictions beyond the first year. There are reasons to think, though, that records play a substantial role in the observed patterns. First, with the exception of felony convictions after their last event, the declines arise immediately in the year of the event without a clear pre-trend. Second, patterns are consistent with the audit literature that finds experimental evidence on reduced call back rates when criminal histories are reported. Third, the effects exhibit similarities to the patterns observed in the survey evidence described above. In particular, the general patterns for first and last events are consistent with our survey evidence that the majority of employers state that they are unwilling to hire someone with even low-level records, that these preferences are stronger for felonies than misdemeanors and for convictions relative to non-convictions, and they are persistent.

These trends suggest that there is potential scope for policies that clear or reduce records to "work" by removing negative information from a person's background check. To the extent the mark itself was a major cause of a sustained decline in employment, removing it could remediate the negative impacts.[26]

IV. Empirical Specification and Analysis of the Fair Credit Reporting Act

For our analysis of the extensive margin of record reporting using records cleared from CRA reports, we define event-time in relation to the year of the charge that gets cleared seven years after the non-conviction or conviction. Our main specification is the same as in Equation 1 except now event time is measured as time since *last* criminal history event. We balance the sample three years prior to the

[26]The patterns also establish confidence in the matching procedure between criminal records and the IRS data given they yield the expected patterns in the data post-event.

clearance event and one year post the event, so that the estimated coefficients around the event are not driven by changes in sample composition. We bin endpoints for the period more than 5 years prior and 10 or more years post event. Our main analysis separately examines "last events" that are felony non-convictions, misdemeanor non-convictions, and felony and misdemeanor convictions. When analyzing convictions, we compare results from Maryland, with the other states without laws governing reporting of convictions.

We use separate analysis samples to analyze the clearance of a non-conviction and conviction. To study non-convictions, we restrict our sample to individuals with a felony or misdemeanor non-conviction as their last event, limited to individuals with no other convictions on their record, because nothing should be reported on these individuals' CRA background checks at the seven year mark, giving FCRA the best chance at improving outcomes for these individuals. To reduce measurement error associated with measuring past convictions, we limit the sample to individuals who were 18 or younger as of the earliest year of data available in the respective jurisdiction to ensure that we can accurately measure their adult criminal record. To study convictions in Maryland versus other jurisdictions in our sample, we restrict our sample to individuals whose last event was either a felony or misdemeanor conviction, regardless of other previous convictions on the record (as those will also be cleared prior to seven years after the last event).

Table 2 presents summary statistics for our analysis samples used in the FCRA and Maryland "last-event" analysis. Columns 1 and 2 present summary statistics for individuals with a felony or misdemeanor non-conviction as their last event, restricted to individuals with no other convictions on their record. Columns 3 and 4 present summary statistics for individuals whose last event was either a felony or misdemeanor conviction, respectively. Summary statistics on baseline outcomes are presented at five years after the charge or disposition date (or alternatively, two years before the record clearance under FCRA or the Maryland law). Among the sample of individuals whose last event is a non-conviction with no other convictions, the average age is approximately 31 years of age, and baseline measures of extensive and intensive employment are generally higher for those whose latest non-conviction was a misdemeanor versus a felony. For example, among last-event misdemeanor non-convictions, 74 percent of individuals had any wages at five years post-charge, with average wages of $19,647. In contrast, among last-event felony non-convictions, 66 percent of individuals had any wages at five years post-charge, with average wages of $14,564. Individuals are older among the sample of people whose last event was a conviction. This sample also has relatively low baseline rates of employment and average wages, particularly among individuals whose last event was a felony conviction.

A. Results

In Figure 5 Panels (a)–(b), we plot event-study coefficients for the removal of a felony and misdemeanor non-conviction, respectively, seven years after the original event. Even though non-conviction events are associated with significant drops in employment rates (as seen in Section III), this figure shows that there is no evidence that removing the last non-conviction from the record of someone with no other convictions increases employment. We find no increases in employment among any group when examining effects separately by race (Appendix Figure A.6).

Figure 5 Panels (c)–(d) shows the same event-study plots for last convictions on record. We see no change in employment rates at the seven year mark in Maryland, where convictions cease to be reported for the majority of individuals in out sample. There is also no divergence in employment rates between Maryland and other states, who do not prohibit the reporting of convictions at a similar event-time.

Tables 3 and 4 report formal tests of whether the event-study coefficients reported in Figure 5 seven and eight years after the last event are different from a linear trend implied by our event-study coefficients, for both any wages and other employment outcomes. We adjust for pre-trends because a criminal history event mechanically occurred seven years before the FCRA event and individuals could still be recovering (or continuing to fall behind) from the initial event.[27] In particular, we observe positive pre-trends in employment for individuals with felony convictions in panel (c) of Figure 5, which are a continuation of the partial recovery of employment rates after a felony conviction observed earlier in panel (c) of Figure 3.[28] Our test evaluates whether trends improve around the FCRA event relative to any pre-existing recovery trend. For exposition purposes, we pool the jurisdictions. Table 3 Panel (a) reports results for misdemeanor non-convictions and Panel (b) reports results for felony non-convictions. Table 4 Panel (a) reports analogous tests of deviations from trend in year 7 and 8 for convictions in the three states that have no state-FCRA law that prohibit the reporting of convictions, and Panel (b) reports results separately for Maryland, which limits the reporting of convictions after 7 years.

Consistent with Panels (a) and (b) of Figure 5, Table 3 shows that for non-convictions (both felony and misdemeanor), we find no detectable deviation from the pre-trend around the timing of the record removal. Given the precision of our estimates, a 95% confidence interval of our any wage outcome rules out employ-

[27] Specifically, we calculate: $d_{+7} = 2 \times \beta_{+4} + \beta_{+7}$ and $d_{+8} = 3 \times \beta_{+4} + \beta_{+8}$ and the standard errors on these sums using the delta method. d_{+7} and d_{+8} report the deviation of our period 7 and 8 event-study coefficients, respectively, from the linear trend implied by our period 4 event-study coefficient, β_{+4}. If β_{+4} is negative, this implies a positive pre-trend. In that case, a positive and statistically significant deviation from trend in periods 7 or 8 would suggest that FCRA is having a positive impact.

[28] The large initial impacts on employment after felony convictions in particular are consistent with transient incapacitation effects from incarceration after felony conviction as documented in Garin et al. (2022).

ment deviations from the pre-trend greater than 0.2 percentage points in year 7 and greater than 0.4 percentage points in year 8. There is similarly no significant discontinuity or deviation from trend for other wage thresholds or employment outcomes.

With respect to convictions, Table 4 Panel (a) shows that individuals appear to be below trend seven to eight years after the initial charge in states (NJ, PA, and TX) that do not prohibit the reporting of convictions, suggesting the positive trend observed closer to the initial event has slowed (see Figure 3 Panels (c) and (d)). Table 4 Panel (b) shows results for convictions in Maryland. We see evidence of positive trends before the event for any wages at various thresholds (see Figure 5 and Appendix Figure A.7). If anything, these trends are *slowing*, not increasing after the conviction is removed at seven years, as indicated by formal tests of deviation from pre-trends.[29] Notably, this pattern in Maryland is similar to the pattern seen for convictions in states that do *not* have a law that removes convictions after seven years. Indeed, 95% confidence intervals rule out *any positive* employment deviations in year 7 and year 8 in the states that do not have a law removing convictions after several years (Panel (a)). And in Maryland, a 95% confidence interval rules out employment deviations greater than 0.3 percentage points in year 7 and greater than 0.4 percentage points in year 8 (Panel (b)).

We see similar trends for tax filing, and little action on our measures of self-employment or independent contracting.[30] In addition, we find no differences by crime type of the last event (see Appendix Figure A.9).

While the removal of these records has no impact on overall employment probabilities or other tax related outcomes, it's possible record removal could impact job switching. Our analysis does not measure job-to-job transitions per se, but we analyze what industry people are working in. To the extent that criminal record remediation helps people access different industries, we might see employment changing across industries at the time of remediation. However, in Figure 6, we can rule out employment increases that deviate from pre-trends across all 2-digit NAICS industries, suggesting that job (or at least industry) switching is also unaffected by the removal of either non-convictions (Panel (a)) or convictions (Panels (b)-(c)).

Overall, the results imply that removing criminal records from CRA background checks after seven years (either non-convictions in NJ, PA, or TX or convictions in MD) does not result in positive impacts on employment and tax-filing outcomes for affected individuals.

Importantly, however, we find evidence that FCRA laws *do* impact employment

[29]Interestingly, defendants identified as Black have positive employment trends while all other races have a downward trend (see Appendix Figure A.6. There is some visual evidence that employment rates rose for Black defendants in Maryland after having their records cleared, but we cannot reject that this increase is simply a continuation of the prior trend.

[30]We find similar null effects at year seven using an alternative differences-in-differences estimator following Sun and Abraham (2021) (see Appendix Figure A.8).

for one particular type of activity: gig economy work mediated by online platforms. Unlike most other types of work where employers consider a broad range of qualifications when making hiring decisions, in most cases platform workers only need to clear one specific bar to participate: a clean criminal record.[31] Thus, policies that remove charges or offenses from one's record should have the clearest impacts on access to such work—which is an important pathway into the labor force for marginally-attached workers.

We present event-study results pooling all criminal history events and jurisdictions are plotted in Figure 7. Because platform-based work has only been prevalent since circa 2012, for this part of our analysis, we restrict our sample to years since 2012 and FCRA "last events" beginning in 2015. First, in the left-hand panel, we examine how platform gig work evolves around one's most recent charge. We find that the time pattern of effects is broadly similar to the effects on traditional employment: gig work declines immediately around the event, suggesting that the presence of a criminal record limits employment opportunities in gig work. However, in sharp contrast to traditional employment, we find that participation in platform gig work increases suddenly and substantially seven years after the criminal history event as FCRA laws begin to apply. We find not only that the negative effects around the "time 0" event are completely offset seven years later, but that the absolute magnitude of the increases exceeds the scale of the initial decline. These findings are consistent with criminal records being the primary barrier to work in this sector and, moreover, suggest that restoring access to this type of work may be particularly important when other work opportunities are limited.

We further examine these results in Column (4) of Tables 3 and 4, which report our test for trend breaks by type of criminal history events for the outcome of any gig work. Interestingly, the aggregated results in the event-study appear to be driven by individuals with convictions rather than those with non-convictions as their latest event. Conviction records are only removed by select state-level FCRA laws as shown in Figure A.2; nonetheless, we find nearly identical 7 effects both for individuals in MD (which has a state FCRA law) and those in state without any law. Importantly, the state of California, where most major gig platform companies are based, has a FCRA law that restricts reporting of convictions after 7 years. Thus, it is likely that platform firms screen primarily based on conviction records and that reporting to such firms adheres to CA state standards.

V. Empirical Specification and Analysis of the Pennsylvania Clean Slate Law

For this analysis we focus on the subset of individuals who only have non-convictions on their records in Pennsylvania, as these individuals' entire criminal

[31] In the period we study, ride-hailing apps account for the overwhelming majority of platform-based gig work observed in the IRS (Garin, Jackson and Koustas, 2022), all of which require drivers to pass a criminal background check.

histories are sealed by PA Clean Slate (if they do not owe fines and fees). We compare the former group to individuals who also only have non-convictions on their records but whose charges were not sealed between June 2019 and June 2020 due to the fines and fees.[32] This comparison allows us to difference out any trends in employment for people with criminal justice system contact during this time period.

Our main specification is given as follows:

$$y_{it} = \beta \text{Cleared}_i \times \mathbb{1}\{Year_t \in \text{2019-2021}\} + \sum_{k \in 2016,2017} \delta_k \text{Cleared}_i \times \mathbb{1}\{Year_t = k\} + X_{it}'\gamma + \alpha_i + \alpha_t + \varepsilon_{it} \quad (2)$$

where Cleared_i is an indicator for an individual having their record sealed (i.e. an ever-treated indicator), and $\mathbf{1}\{Year_t \in \text{2019-2021}\}$ is an indicator for being in the period after records were cleared. The interpretation of β is the difference-in-differences estimator, comparing the change in the outcome in the post period between those that had all their misdemeanor charges sealed with those who did not because they owed fines and fees. The interactions between Cleared_i and earlier years provide a test for pre-trends. We use data since 2016 for this analysis and restrict our analysis sample to individuals aged 18 to 25 to ensure that they had no other prior convictions by the start of the PA charge data, which begins in 2008.

In contrast to FCRA and the Maryland Credit law, the Pennsylvania law allows us to estimate the effect of sealing non-conviction records that are more recent than seven years from the initial charge. To examine if the effect varies with the time elapsed since the last charge, we also estimate a triple-difference specification including additional interactions with months since the latest charge (calculated as months since June 2019):

$$y_{it} = \beta_1 \text{Cleared}_i \times \mathbb{1}\{Year_t \in \text{2019-2021}\} + \beta_2 \mathbb{1}\{Year_t \in \text{2019-2020}\} \times \text{Months since charge}_i + \beta_3 \text{Cleared}_i \times \mathbb{1}\{Year_t \in \text{2019-2021}\} \times \text{Months since charge}_i + \sum_{k \in 2016,2017} \delta_k \text{Cleared}_i \times \mathbb{1}\{Year_t = k\} + X_{it}'\gamma + \alpha_i + \alpha_t + \varepsilon_{it} \quad (3)$$

The interpretation of the coefficient on $\text{Cleared}_i \times \mathbf{1}\{Year_t \in \text{2019-2021}\}$ is now the out-of-sample predicted effect at zero months since the initial charge. The earliest sealings we are able to see in our data occur approximately 18 months after the charge, as we have data through 2018 and the sealings took place June

[32] We do not know the exact date of sealing for each case, only whether it was sealed between June 2019 and June 2020.

2019–June 2020.

Summary statistics for our estimation sample are presented in Table 5. We break these out by those who are cleared in 2019-2020 ("Treated") and those who did not because they owed fines/fees ("Control.") While the two groups are broadly similar, the treated are less likely to be male, more likely to be Black, and had somewhat older charges. They are also slightly less likely to be employed, although the probability of making $15,000 per year (approximately a full-time/full-year job at the minimum wage) and average wages (including 0s) are similar across the two groups.

A. Results

Figure 8 panel (a) and panel (c) show raw trends in having any employment, or any employment earning more than $15,000 per year, over our time period, by treatment and control groups. Panels (b) and (d) report the event-study estimates. While both the treatment and control groups were hit hard during COVID, we do not see any differential trends among those with a cleared record.

Table 6 Panel (a) presents the main pooled differences-in-differences results and Panel (b) presents estimates from our triple-differences specification to test whether more recently-sealed charges have a differential effect on employment. While these non-conviction records were generally sealed less than seven years before the original event (on average sealed 5-6 years after the charge), we find no detectable effect of the automated record clearance for treated versus control individuals for a range of employment outcomes, although we do find a marginally significant effect for gig platform work for those with more recent charges.[33] Of course, these automated sealings took place at the beginning of the COVID-19 pandemic, making it difficult to draw conclusions about the potential impacts of sealings in a non-pandemic period. Appendix Tables A.5-A.6 shows results by race. While the overall effects are similar, there is some suggestive evidence that clearing the records earlier is beneficial for Blacks.

Overall, these findings suggest that full and automated sealing of non-convictions from criminal records does not result in improved employment outcomes, even for those with relatively recent non-convictions that are around 18 months old.

VI. Discussion and Conclusion

In this article, we study the impact of criminal record remediation policies. With a few exceptions, these policies did not improve the labor market prospects of individuals with records, on average, despite their great promise.

[33]To ensure our results are not driven by Philadelphia, which generally does not levy fines and fees for non-convictions and thus had very few defendants in the control group, in Appendix Table A.4 we exclude Philadelphia and find similar results, although the result for gig platform work is attenuated from removing this large urban county where gig work would be concentrated.

There could be a myriad of factors for why these policies are not producing their intended results. If lower level convictions or non-convictions are not associated with declines in employment when they occur, then there is no reason to believe that remediating them later would help. But we provide evidence of large and persistent declines in employment around the time of both first and last charges, even those that led to non-convictions. We also provide evidence that many employers perform background checks and that most use a CRA, with less than 10% saying they do background checks "in-house". In other work, we also found that a policy to reduce past felonies to misdemeanors had no statistically significant impact on employment for those whose reductions were done exogenously by public defender without their input (Agan et al. 2024). Like the policies we study here, this could have been because the recipients of the remediation did not know; however an RCT to notify these individuals did not improve outcomes for the notification group over the control, which may imply limited scope for lack of knowledge to be driving our results (Agan et al. 2024). Another explanation for the null result might be that employers are asking applicants about records, such that any information obtained in a background check is superfluous. To examine this mechanism, we estimate the main FCRA specifications for New Jersey after the state Ban the Box (BTB) law went into effect in March 2015. Appendix Figure A.10 shows that even in post-BTB years, there remains a null effect after seven years of record sealing.

Of course we cannot rule out other possible mechanisms for our findings. Criminal records leave other types of digital footprints that can be hard to erase (Lageson 2020) and if employers are searching for these records they may, in essence, "undo" the remediation. However, a large majority of employers would need to be undertaking these sorts of costly searches to fully explain our null results, and they would need to turn up criminal history information for a large number of applicants. For larger employers this would be highly impractical. To further check whether this could explain our null results, for a random sample of a little over 200 defendants in our data, we had undergraduate RAs perform internet search for these defendants and see if they found any concerning information about crimes - but they only found crime information for 20% of defendants which is unlikely to explain our null result.[34].

The findings point to the conclusion that older records do not have an independent penalty on employment for most jobs. Our survey results suggest that this could be, in part, because employers largely are willing to overlook older records. However, this fact alone would suggest that the employment impacts of a record should be short-lived, whereas we observe persistent negative impacts following both charges that lead to convictions and non-convictions. A more complete account is that charges can lead to scarring such as resume gaps, loss of experience, discouragement, and reduced search (Smith and Broege 2019) that can be difficult to undo. Meanwhile, our results highlight that in specific sectors like platform-

[34]Further details about these searches and this data can be found in Appendix E

based gig work, which are open to all workers regardless of other qualifications *but* for which criminal records themselves have direct impacts on access, record remediation policies can have immediate effect on job access.

The scarring mechanism is interesting in light of the sectoral shifts that we observe post-charge. For both non-convictions and convictions there is a relative shift towards interim type jobs, namely temp agencies, retail and food and accommodations. In an audit study, Farber et al. (2019) find significantly lower call-back rates for workers in interim jobs. The evidence suggests that charges may lead to changes in labor market outcomes from path-dependence due both to loss of employment and a move to sectors that result in permanently reduced job finding rates.

We also find large increases in the probability an individual has earnings from a gig platform after remediation. These results for gig platform work provide insight into the scarring mechanism since a record can be the principal barrier to employment in this line of work. Our findings on gig work suggest that non-traditional work arrangements, which have typically been unstudied, may be an increasingly-important avenue of work for criminal justice-involved individuals.

If well-intended record remediation initiatives do not improve labor market outcomes, what policies might work? Experimental evidence on well-known programs that provide transitional jobs or mentoring and work readiness training for those recently released from prison suggest that these initiatives have also been largely ineffective at improving employment outcomes. For example, the Reintegration of Ex-Offenders (RExO) project, a joint initiative between various federal agencies including the Department of Labor and Department of Justice, provides individuals with records with a range of services: mentoring, work readiness training, and job placement, among others. Yet, a randomized evaluation of RExO found no evidence of positive impacts of the program on either recidivism or labor market outcomes within three years of program enrollment. Qualitative evidence from the study suggested that a comprehensive and intensive approach that helps address the wide array of other issues present in the ex-offender population (housing, substance abuse etc.) may be needed (Wiegand and Sussell 2016). Similarly, a randomized evaluation of the New York City-based Center for Employment Opportunities (CEO) program, which provides transitional employment and other supportive services for recently released individuals, was also largely ineffective at increasing longer-term employment (Valentine and Redcross 2015).

Our findings, coupled with this literature, suggest that policies that clear records earlier, or which offer more intensive wrap-around programs that actively seek to connect those with records to stable long-term employment, may generate larger labor market impacts. The evidence from our research suggests that scarring from records can occur quickly, consistent with Agan, Doleac and Harvey (2023) who find that negative impacts of a misdemeanor charge on recidivism occurs within the first 15 months. As a result, automated clean slate policies with shorter wait-

ing periods, such as immediate sealing of non-convictions and convictions, could be more fruitful. We could not study immediate sealings in Pennsylvania due to data limitations.

Of course, any benefits from a policy that shortens the reporting period on records would have to be weighed against potential costs. Records could provide information that may be useful to employers given concerns about risk and productivity (Cullen, Dobbie and Hoffman, 2023). For example, using our data from Bexar County we find that having a recent record, even a non-conviction, is highly predictive of having another criminal event occur in the following five years.[35] Interestingly, by six years after their last event, people who had been convicted of a crime or been charged but not convicted have the same likelihood of committing a crime as a random person in the population. Future research is needed to determine whether the benefits of earlier record remediation policies outweigh the potential costs for employers. Given the persistent labor market impacts following criminal events, which are not undone by record sealing several years later, developing and evaluating remediation policies with shorter waiting periods, such as an evaluation of immediate non-conviction sealing under Pennsylvania Clean Slate, is an important area for future work.

[35]See Appendix Figure A.11.

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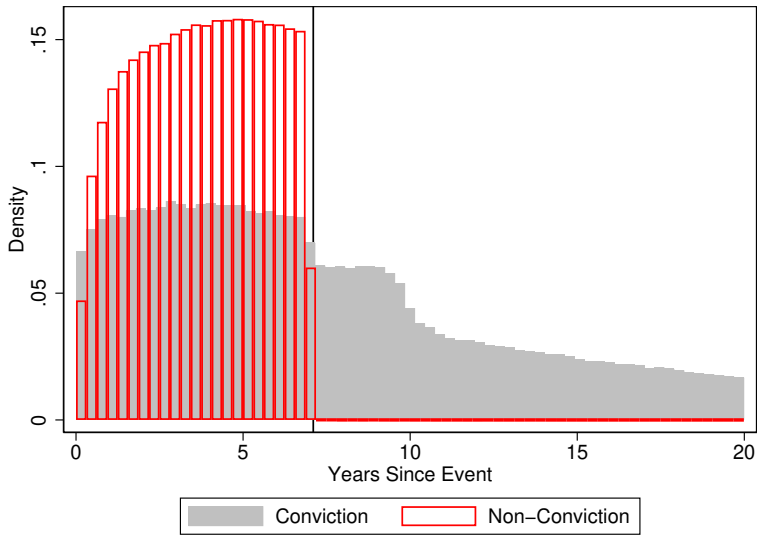
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Figures and Tables

Figures

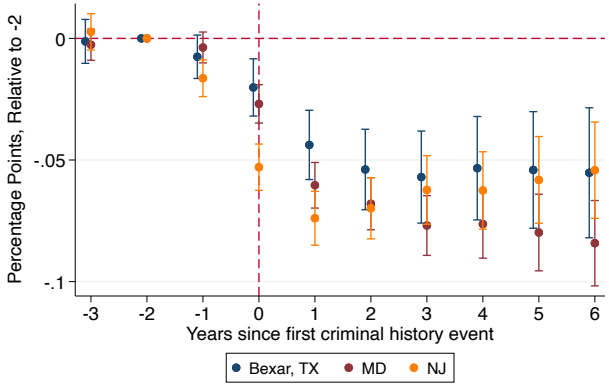
Figure 1. Years Elapsed for Charges Reported on Criminal Background Checks



Notes: Data obtained from Agan, Autor and Rackstraw (2024). Includes information on all criminal background checks performed by a large CRA for 228 firms. These background checks include > 11 million charges reported to employers. 28% of charges reported were dismissed charges, 60% conviction charges, and 12% other. This figure reports the distribution of time since event (charge for non-conviction, conviction date for convictions) for the dismissed and conviction charges.

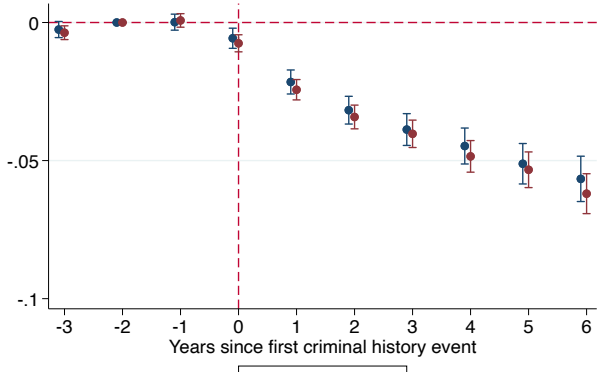
Figure 2. Persistent reductions in employment after *initial* charge

(a) Felony Non-Convictions



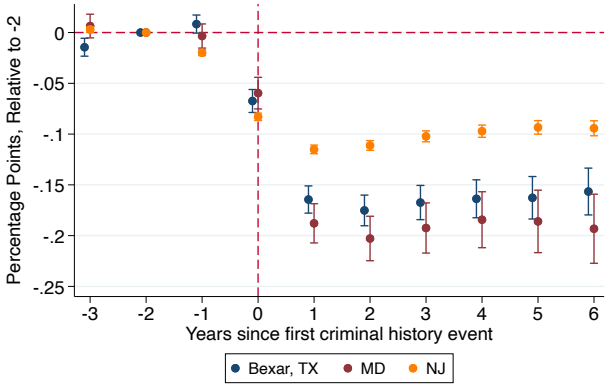
Bexar, TX: N= 7,363, NxT= 142,661, Dep. Mean in -2: 0.763
MD: N= 16,781, NxT= 285,610, Dep. Mean in -2: 0.726
NJ: N= 13,762, NxT= 251,551, Dep. Mean in -2: 0.686

(b) Misdemeanor Non-Convictions



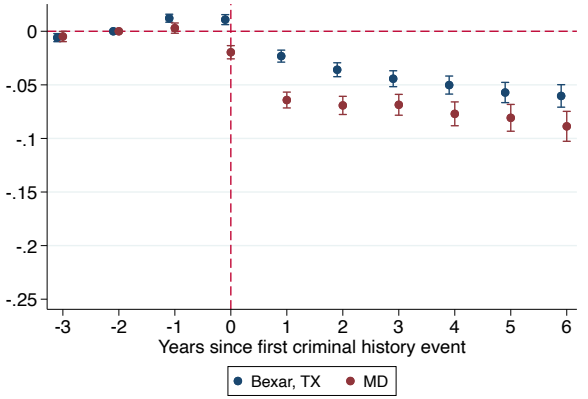
Bexar, TX: N= 59,737, NxT=1,140,081, Dep. Mean in -2: 0.800
MD: N= 100,356, NxT=1,741,007, Dep. Mean in -2: 0.749

(c) Felony Convictions



Bexar, TX: N= 9,885, NxT= 192,419, Dep. Mean in -2: 0.709
MD: N= 5,238, NxT= 92,699, Dep. Mean in -2: 0.729
NJ: N= 99,228, NxT=1,820,135, Dep. Mean in -2: 0.685

(d) Misdemeanor Convictions

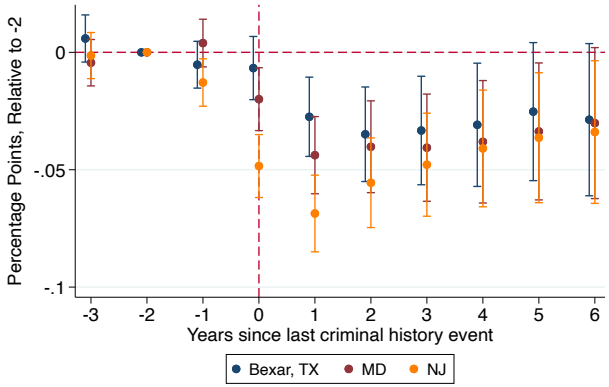


Bexar, TX: N= 43,100, NxT= 824,407, Dep. Mean in -2: 0.770
MD: N= 28,334, NxT= 506,063, Dep. Mean in -2: 0.736

Notes: Each panel plots selected event study coefficients for the share with any wages around an individual's first charge, following specification 1 in the text. For this analysis, we restrict the full sample to have been 18 by the time the first charge appears in the data for both non-convictions and convictions. Coefficients are relative to -2 periods before the first charge. We run separate event studies for each state in each panel. Data from 2000-2020. The sample is restricted to events occurring between 1996-2011 to ensure the regression is balanced in the event window.

Figure 3. Reductions in Employment After Latest Event

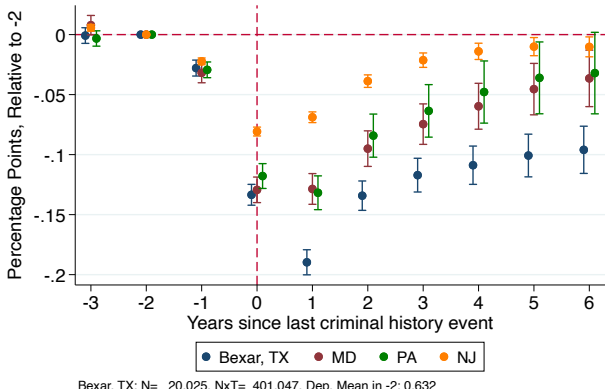
(a) Fel. Non-Convictions, no earlier convictions



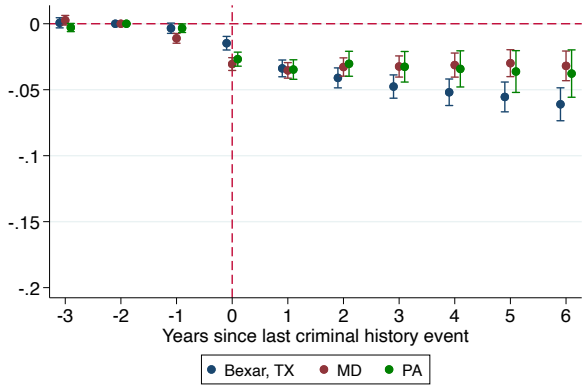
(b) Mis. Non-Convictions, no earlier convictions



(c) Fel. Convictions



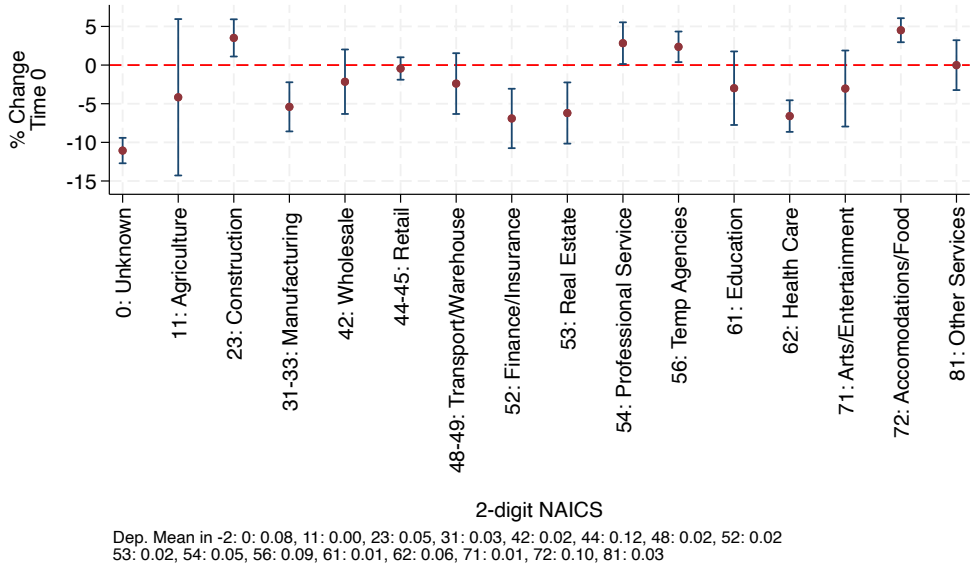
(d) Mis. Convictions



Notes: Each panel plots selected event study coefficients for the share with any wages > $0 around the latest event, following specification 1 in the text. This event is the charge date for non-convictions and disposition date for convictions. Coefficients are relative to -2 years before the event. We run separate event studies for each state in each panel. Data from 2000-2020. The sample is restricted to events occurring between 2003-2011.

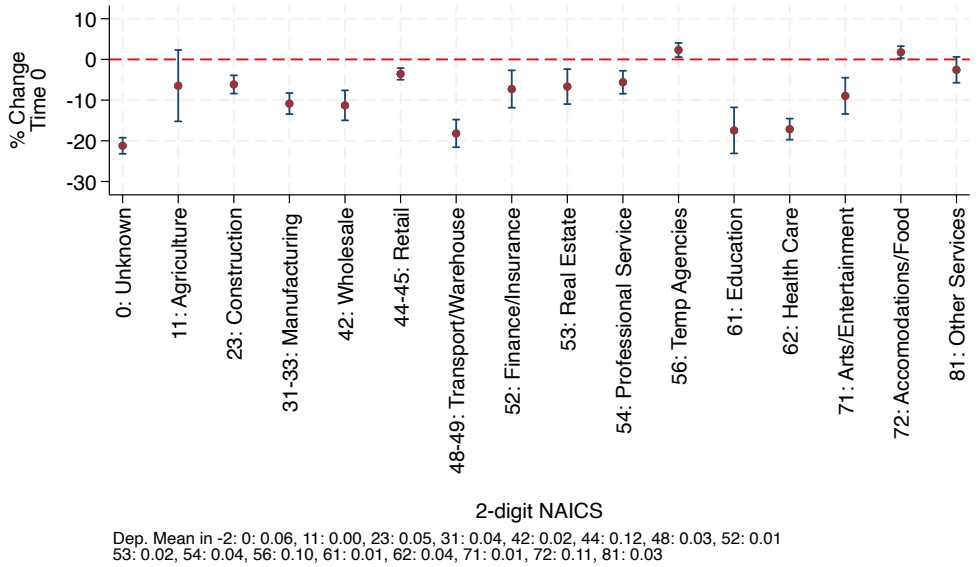
Figure 4. Event Study of Any Wages Around First Criminal History Event, By 2-digit NAICS Industry

(a) Non-Convictions



Dep. Mean in -2: 0: 0.08, 11: 0.00, 23: 0.05, 31: 0.03, 42: 0.02, 44: 0.12, 48: 0.02, 52: 0.02
53: 0.02, 54: 0.05, 56: 0.09, 61: 0.01, 62: 0.06, 71: 0.01, 72: 0.10, 81: 0.03

(b) Convictions



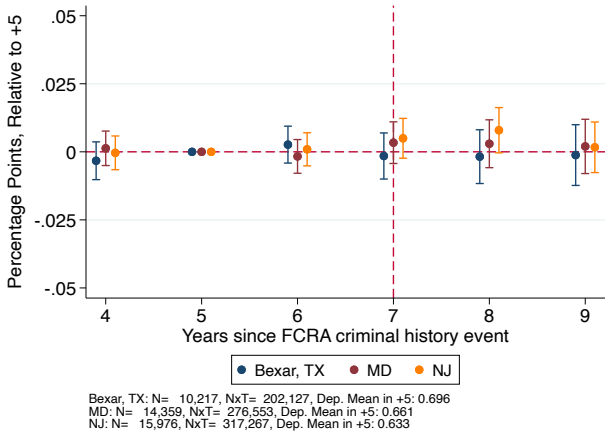
Dep. Mean in -2: 0: 0.06, 11: 0.00, 23: 0.05, 31: 0.04, 42: 0.02, 44: 0.12, 48: 0.03, 52: 0.01
53: 0.02, 54: 0.04, 56: 0.10, 61: 0.01, 62: 0.04, 71: 0.01, 72: 0.11, 81: 0.03

Notes: Figure reports event study estimates in the year of someone's first criminal history event of a W-2 issued by a payer firm in the specified 2-digit NAICS code based on the firms' tax return in that year. For this analysis, we restrict the full sample to have been 18 by the time the first charge appears in the data for both non-convictions and conviction. Data from 2000-2020. The sample is restricted to events occurring between 2003-2018. We run separate event studies by industry around the criminal history event, and divide by the mean share in the in the industry in -2 and multiply by 100 to convert to a percent change.

Figure 5. FCRA Event Study of Any Wages Around Removal (Year 7)

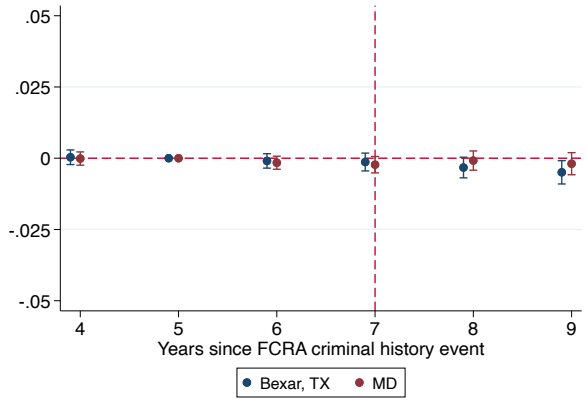
Note: MD has State FCRA for Convictions

(a) Felony Non-Convictions, no other convictions



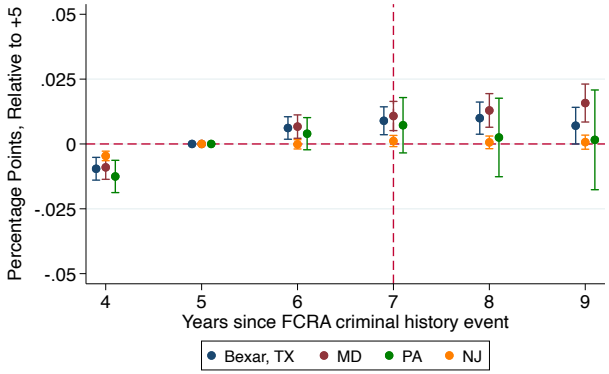
Bexar, TX: N= 10,217, NxT= 202,127, Dep. Mean in +5: 0.696
MD: N= 14,359, NxT= 276,553, Dep. Mean in +5: 0.661
NJ: N= 15,976, NxT= 317,267, Dep. Mean in +5: 0.633

(b) Mis. Non-Convictions, no other convictions



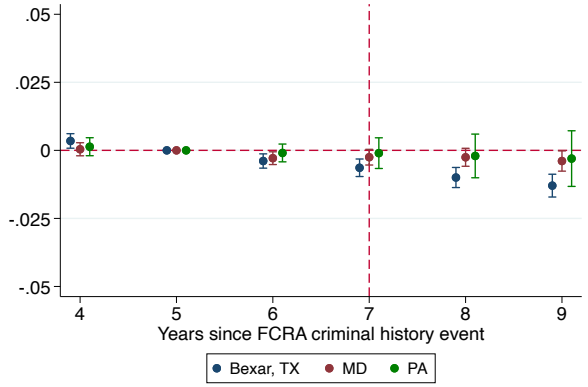
Bexar, TX: N= 68,694, NxT=1,345,682, Dep. Mean in +5: 0.754
MD: N= 90,519, NxT=1,722,568, Dep. Mean in +5: 0.747

(c) Felony Convictions



Bexar, TX: N= 28,786, NxT= 576,704, Dep. Mean in +5: 0.480
MD: N= 23,719, NxT= 473,663, Dep. Mean in +5: 0.465
PA: N= 35,163, NxT= 676,163, Dep. Mean in +5: 0.483
NJ: N= 194,068, NxT=3,894,497, Dep. Mean in +5: 0.547

(d) Mis. Convictions

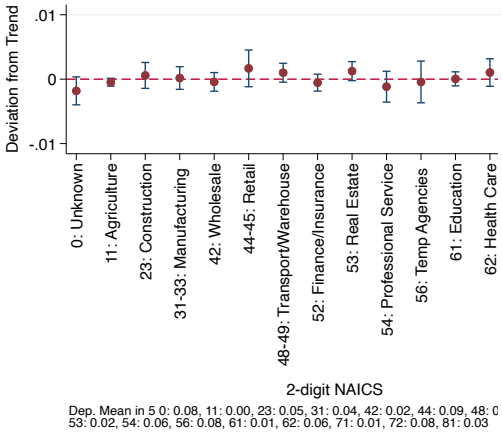


Bexar, TX: N= 68,336, NxT=1,362,099, Dep. Mean in +5: 0.643
MD: N= 86,508, NxT=1,733,092, Dep. Mean in +5: 0.569
PA: N= 106,080, NxT=2,073,847, Dep. Mean in +5: 0.618

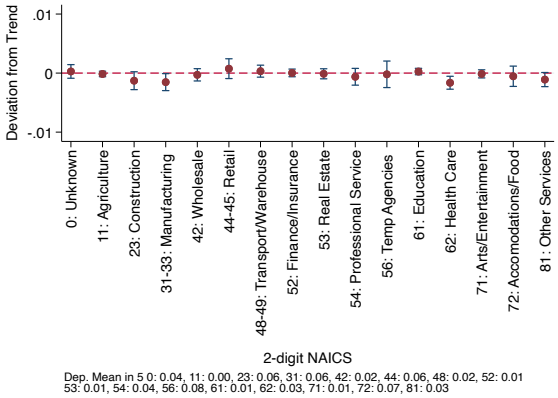
Notes: Each panel plots selected event study coefficients for the share with any wages > $0 around 7 years after the event, following specification 1 in the text. Timing from the event is based on the charge date for non-convictions and disposition date for convictions. Coefficients are relative to +5 periods after the event (2 years prior to the year 7 FCRA event, if applicable). We run separate event studies for each state in each panel. Data from 2000-2020. The sample is restricted to events occurring between 1996-2011 to ensure the regression is balanced in the event window.

Figure 6. FCRA Event Study of Any Wages Around Removal (Year 7), Deviation from Trend, By 2-digit NAICS Industry

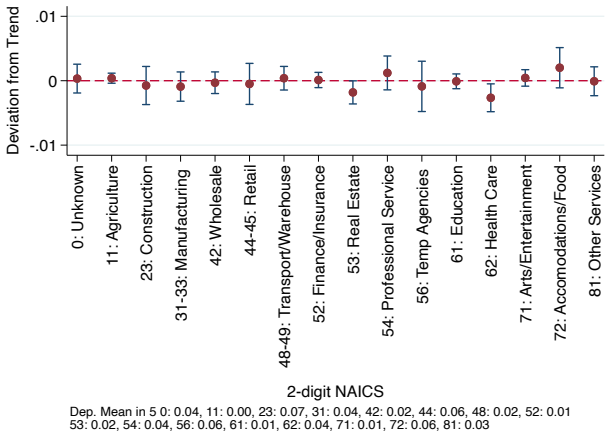
(a) Non-Convictions



(b) Convictions in PA, NJ and TX

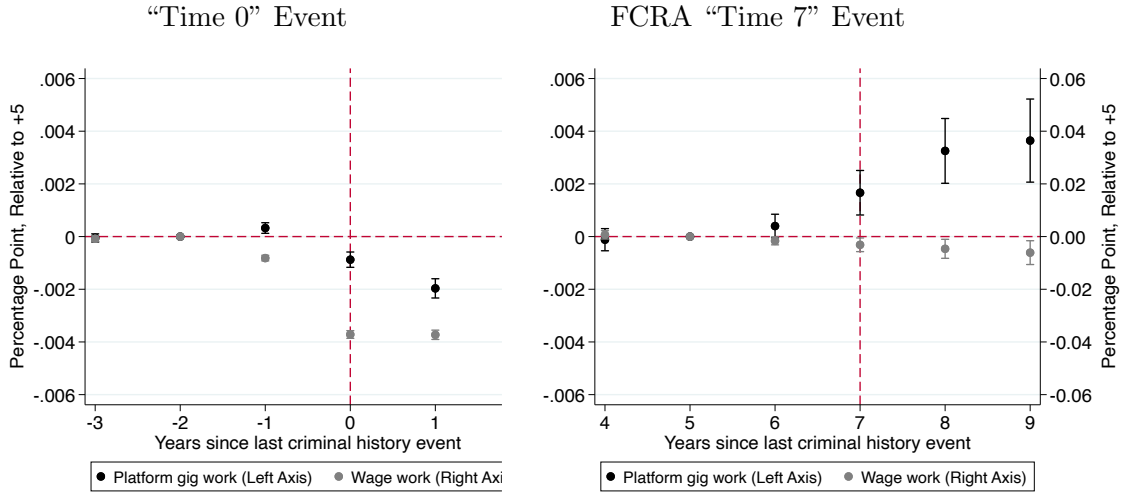


(c) Convictions in MD



Notes: Figure reports results from a test of whether the event study coefficients 7 years after the last charge are different from a linear trend. Specifically, figure reports $2 \times \beta_{+4} + \beta_{+7}$.

Figure 7. FCRA Event-Study Estimates: Any Gig Platform Work

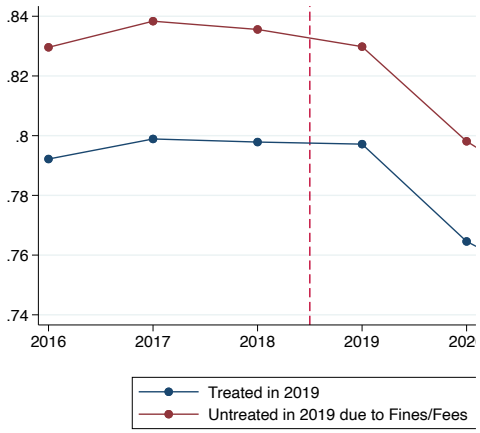


Notes: Each panel plots selected event study coefficients for the share with any gig income > \$0 or wage income > \$0 in different windows around the latest event. Events are defined as charge dates for non-convictions and disposition dates for convictions. The left panel is restricted to latest events from 2015-2018 and focuses on the period around the initial event. The right panel is restricted to latest events from 2008-2011 and focuses on the period 7 years after the latest event. The data are restricted to the period since 2012. We pool all charges for this figure.

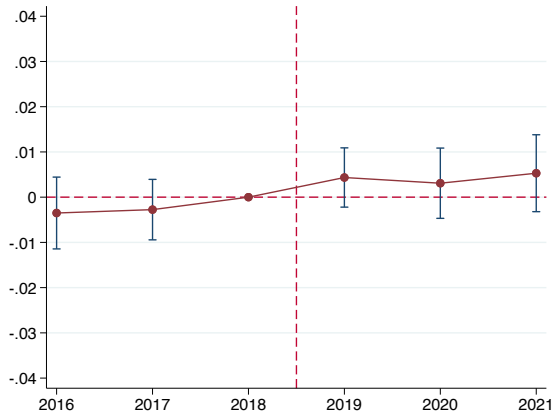
Figure 8. Impact of PA Clean Slate Reductions on Employment Outcomes

Any Wages > $0

(a) Raw Data

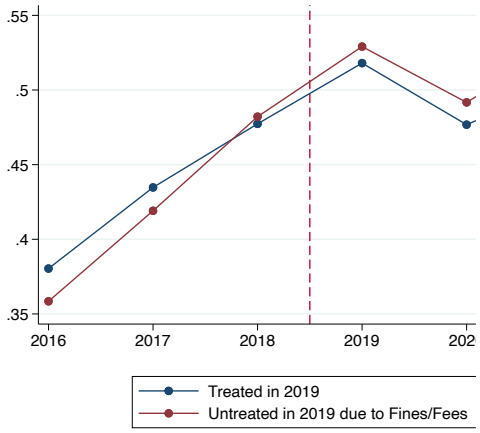


(b) Event-Study Estimates

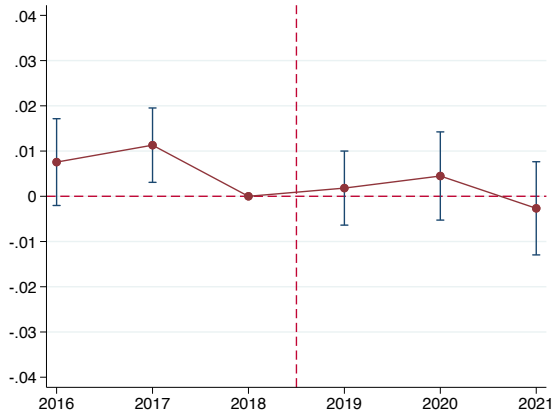


Any Wages > $15,000

(c) Raw Data



(d) Event-Study Estimates



Notes: Figure reports raw means and event-study estimates for those who had their non-convictions cleared by PA's Clean Slate law by 2020, compared with those who did not. Data from 2016-2021. Sample is restricted to ages 18-25 to ensure they had no other prior convictions by the start of our charge data, which begins in 2008.

Tables

Table 1— Survey of Hiring Professionals on Willingness to Hire Misdemeanor vs. Felony Convictions

	Drug Conviction			Theft Conviction		
	Misd	Felony	Misd-Felony	Misd	Felony	Misd-Felony
Prob Would Hire if Crime Was...						
1-3 Years Ago	0.42	0.29	0.13**	0.28	0.14	0.14***
			(0.05)			(0.04)
4-7 Years Ago Years Ago	0.75	0.63	0.12**	0.61	0.51	0.10*
			(0.05)			(0.05)
8-10 Years Ago	0.91	0.83	0.08**	0.82	0.76	0.06
			(0.04)			(0.05)
N	121	134		125	128	
	Drug Non-Conviction			Theft Non-Conviction		
	Misd	Felony	Misd-Felony	Misd	Felony	Misd-Felony
Prob Would Hire if Crime Was...						
1-3 Years Ago	0.60	0.50	0.10*	0.46	0.42	0.04
			(0.05)			(0.04)
4-7 Years Ago Years Ago	0.82	0.81	0.01	0.73	0.78	-0.05
			(0.05)			(0.05)
8-10 Years Ago	0.88	0.88	0.00	0.86	0.90	-0.04
			(0.04)			(0.05)
N	116	124		111	111	

Notes: Survey of 1003 hiring professionals with experience in the United States in the past 5 years. Each respondent was randomly assigned to being asked about preferences for hiring someone with one of 8 potential criminal histories: (drug x theft) + (misd x felony) + (conviction x non-conviction). The question text was "You intend to hire a candidate for an open entry-level position at the most recent firm at which you had hiring experience. Through the hiring process, you decide that this candidate is well qualified for the position. You are ready to extend an offer to the candidate. However, you learn that the candidate was charged with [**crime type**] [**X years ago**] and was convicted. How likely are you to recommend that the company hire the candidate?" Choices were: Definitely will, probability will, probability will not, definitely will not. The respondent was asked this question for X from 1 through 10 on the same page. This table combines "Definitely will" and "probably will," and shows the average probability the respondent reported would hire in bins of years.

Table 2— Summary Statistics: Last-Event Analysis Estimation Sample

	(1)	(2)	(3)	(4)
	Last Event is Non-Conv & No Other Conv		Last Event is Conv.	
	Felony	Misdemeanor	Felony	Misdemeanor
Male	0.648	0.643	0.793	0.765
5 Years After Charge/Disposition:				
Age	30.95	30.61	38.18	39.69
Any Wages	0.659	0.743	0.525	0.608
Wages>$15k	0.390	0.499	0.279	0.399
Avg. Wages	14,564	19,647	10,803	16,533
Any 1099 NEC	0.067	0.086	0.053	0.079
Filed Taxes	0.639	0.739	0.479	0.589
Any SE Income	0.073	0.078	0.049	0.061
Total Obs	40,552	171,725	280,854	260,515

Notes: Uses charges in the case of non-convictions, or disposition dates in the case of convictions. Sample is restricted to charges or dispositions from 1996 to 2011.

Table 3— Test for Deviations from Trend Around "Year 7" FCRA Event

(a) Mis. non-convictions and no other convictions, MD and Bexar, TX

	(1) Any Wages>$0	(2) ...>$7,500	(3) ...>$15,000	(4) Any Gig	(5) Any Other 1099	(6) Files 1040	(7) Files SE
+7 Trend Deviation	-0.002	-0.004[a]	-0.004	-0.001	0.004*	-0.001	0.001
(S.E.)	(0.002)	(0.003)	(0.003)	(0.001)	(0.002)	(0.002)	(0.002)
+8 Trend Deviation	-0.002	-0.007[a]	-0.004	-0.001	0.006*	-0.000	0.002
(S.E.)	(0.003)	(0.003)	(0.003)	(0.002)	(0.003)	(0.003)	(0.002)
N	159,194	159,194	159,194	57,246	159,194	159,194	159,194
NxT	3,068,250	3,068,250	3,068,250	515,235	3,068,250	3,068,250	3,068,250

(b) Felony non-convictions and no other convictions, MD, NJ, and Bexar, TX

	(1) Any Wages>$0	(2) ...>$7,500	(3) ...>$15,000	(4) Any Gig	(5) Any Other 1099	(6) Files 1040	(7) Files SE
+7 Trend Deviation	0.001	0.006	-0.009[a]	-0.001	0.006[a]	-0.004	0.001
(S.E.)	(0.005)	(0.005)	(0.005)	(0.002)	(0.003)	(0.005)	(0.003)
+8 Trend Deviation	0.002	0.008	-0.013[a]	0.000	0.006[a]	-0.003	-0.000
(S.E.)	(0.007)	(0.007)	(0.007)	(0.003)	(0.005)	(0.007)	(0.004)
N	40,540	40,540	40,540	13,480	40,540	40,540	40,540
NxT	795,947	795,947	795,947	121,324	795,947	795,947	795,947

Notes: Table reports results from a test of whether the event study coefficients 7-8 years after the last charge are different from a linear trend. Specifically, "+7 Trend Deviation" reports $2 \times \beta_{+4} + \beta_{+7}$, and "+8 Trend Deviation" reports $3 \times \beta_{+4} + \beta_{+8}$. With the exception of Column (4), data is from 2000-2020, and the sample is restricted to events occurring between 1996-2011 to ensure the regression is balanced in the window around the event. Column (4) restricts to data from 2012 and events from 2015-2018. The outcomes in Columns 1-3 are indicators for total wage and salary earnings reported on W-2 returns exceeding the stated treshhold. The outcome in Column 4 is an indicator for having any payments reported by an online platform company on form 1099-MISC or 1099-K. The outcome in Column 5 is an indicator for having any other non-employee compensation (i.e. contractor) payments reported by firms on form 1099-MISC. The outcomes in Columns 6 and 7 are indicators for whether the individual filed an income tax return and whether they reported self-employment earnings on that return, respectively. Standard errors clustered on individual are reported in parentheses. [a] p<0.1, * p<0.05, ** p<0.01, *** p<0.001

Table 4— Test for Deviations from Trend Around "Year 7" in MD v All Other States

(a) Convictions in NJ, PA and Bexar, TX [No state FCRA]

	(1) Any Wages>$0	(2) ...>$7,500	(3) ...>$15,000	(4) Any Gig	(5) Any Other 1099	(6) Files 1040	(7) Files SE
+7 Trend Deviation	-0.006***	-0.006***	-0.003*	0.002***	-0.001	-0.010***	-0.001
(S.E.)	(0.002)	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.001)
+8 Trend Deviation	-0.011***	-0.010***	-0.006*	0.004***	-0.002	-0.012***	-0.001
(S.E.)	(0.002)	(0.002)	(0.002)	(0.000)	(0.001)	(0.002)	(0.001)
N	430,324	430,324	430,324	227,737	430,324	430,324	430,324
NxT	8,584,190	8,584,190	8,584,190	2,050,582	8,584,190	8,584,190	8,584,190

(b) Convictions in MD [State FCRA]

	(1) Any Wages>$0	(2) ...>$7,500	(3) ...>$15,000	(4) Any Gig	(5) Any Other 1099	(6) Files 1040	(7) Files SE
+7 Trend Deviation	-0.003	-0.008**	-0.010***	0.002	0.000	-0.008**	0.001
(S.E.)	(0.003)	(0.003)	(0.003)	(0.001)	(0.002)	(0.003)	(0.002)
+8 Trend Deviation	-0.004	-0.012**	-0.015***	0.004**	0.002	-0.009**	0.000
(S.E.)	(0.004)	(0.004)	(0.004)	(0.001)	(0.003)	(0.004)	(0.002)
N	110,227	110,227	110,227	34,881	110,227	110,227	110,227
NxT	2,206,755	2,206,755	2,206,755	312,769	2,206,755	2,206,755	2,206,755

Notes: Table reports results from a test of whether the event study coefficients 7-8 years after the disposition are different from a linear trend. Specifically, "+7 Trend Deviation" reports $2 \times \beta_{+4} + \beta_{+7}$, and "+8 Trend Deviation" reports $3 \times \beta_{+4} + \beta_{+8}$. With the exception of column (4), data is from 2000-2020, and the sample is restricted to events occurring between 1996-2011 to ensure the regression is balanced in the window around the event. Column (4) restricts to data from 2012 and events from 2015-2018. The outcomes in Columns 1-3 are indicators for total wage and salary earnings reported on W-2 returns exceeding the stated treshhold. The outcome in Column 4 is an indicator for having any payments reported by an online platform company on form 1099-MISC or 1099-K. The outcome in Column 5 is an indicator for having any other non-employee compensation (i.e. contractor) payments reported by firms on form 1099-MISC. The outcomes in Columns 6 and 7 are indicators for whether the individual filed an income tax return and whether they reported self-employment earnings on that return, respectively. Standard errors clustered on individual are reported in parentheses. [a] p<0.1, * p<0.05, ** p<0.01, *** p<0.001

Table 5— PA Clean Slate Estimation Sample, Summary Statistics

	Pooled	Treated	Control
Male	0.645	0.631	0.682
Black	0.303	0.314	0.276
Age	26.39	26.77	25.45
Years Since Charge	4.781	5.122	3.941
Any Wages	0.809	0.798	0.835
Wages>$15k	0.480	0.478	0.484
Avg. Wages	19,531	19,566	19,445
Any 1099 NEC, non platform	0.056	0.056	0.056
Any platform gig	0.011	0.011	0.010
Filed Taxes	0.713	0.706	0.731
Any SE Income	0.051	0.053	0.047
Total Obs	45,877	32,677	13,221

Notes: Summary statistics for PA estimation sample. This sample had charges that were eligible to be dismissed or withdrawn, and were 18-25 at time of the charge.

Table 6— Impact of PA Clean Slate Reductions on Employment Outcomes

(a) DD Estimates

	(1) Any Wages>$0	(2) ...>$7,500	(3) ...>$15,000	(4) Any Gig	(5) Any Other 1099	(6) Files 1040	(7) Files SE
Post (2019-2021) × Cleared	0.004	-0.001	0.001	0.001	0.001	0.008[a]	-0.004[a]
	(0.003)	(0.004)	(0.004)	(0.001)	(0.002)	(0.004)	(0.002)
2017 × Cleared	-0.003	-0.001	0.011**	-0.001	-0.007**	-0.006	-0.005*
	(0.003)	(0.004)	(0.004)	(0.001)	(0.002)	(0.004)	(0.002)
2016 × Cleared	-0.004	-0.001	0.008	0.002	-0.004	-0.005	-0.002
	(0.004)	(0.005)	(0.005)	(0.001)	(0.003)	(0.005)	(0.002)
Dep. Mean (2018)	0.809	0.618	0.480	0.011	0.056	0.713	0.051
N	45,877	45,877	45,877	45,877	45,877	45,877	45,877
NxT	275,634	275,634	275,634	275,634	275,634	275,634	275,634
Age Controls	X	X	X	X	X	X	X
Indiv. FE	X	X	X	X	X	X	X
Year FE	X	X	X	X	X	X	X

(b) By months since charge

	(1) Any Wages>$0	(2) ...>$7,500	(3) ...>$15,000	(4) Any Gig	(5) Any Other 1099	(6) Files 1040	(7) Files SE
Post (2019-2021) × Cleared	0.002	-0.008	-0.023*	0.007*	-0.003	0.021*	-0.002
	(0.008)	(0.010)	(0.010)	(0.004)	(0.005)	(0.009)	(0.004)
Post (2019-2021) × Cleared × Months since charge	-0.000	0.000	0.000**	-0.000*	0.000	-0.000	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Post (2019-2021) × Months since charge	0.000**	-0.000	-0.000***	0.000	-0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
2017 × Cleared	-0.003	-0.000	0.012**	-0.001	-0.007**	-0.006	-0.005*
	(0.003)	(0.004)	(0.004)	(0.001)	(0.002)	(0.004)	(0.002)
2016 × Cleared	-0.005	-0.000	0.008[a]	0.002	-0.004	-0.005	-0.002
	(0.004)	(0.005)	(0.005)	(0.001)	(0.003)	(0.005)	(0.002)

Notes: Table reports difference-in-differences results comparing outcomes for individuals who had all their non-convictions cleared by PA's Clean Slate law by 2020, compared with those who did not. Data from 2016-2021. Sample is restricted to ages 18-25 to ensure they had no other prior convictions by the start of our charge data, which begins in 2008. Outcomes are as described in notes to Table 3. Standard errors clustered on individual are reported in parentheses. [a] p<0.1, * p<0.05, ** p<0.01, *** p<0.001